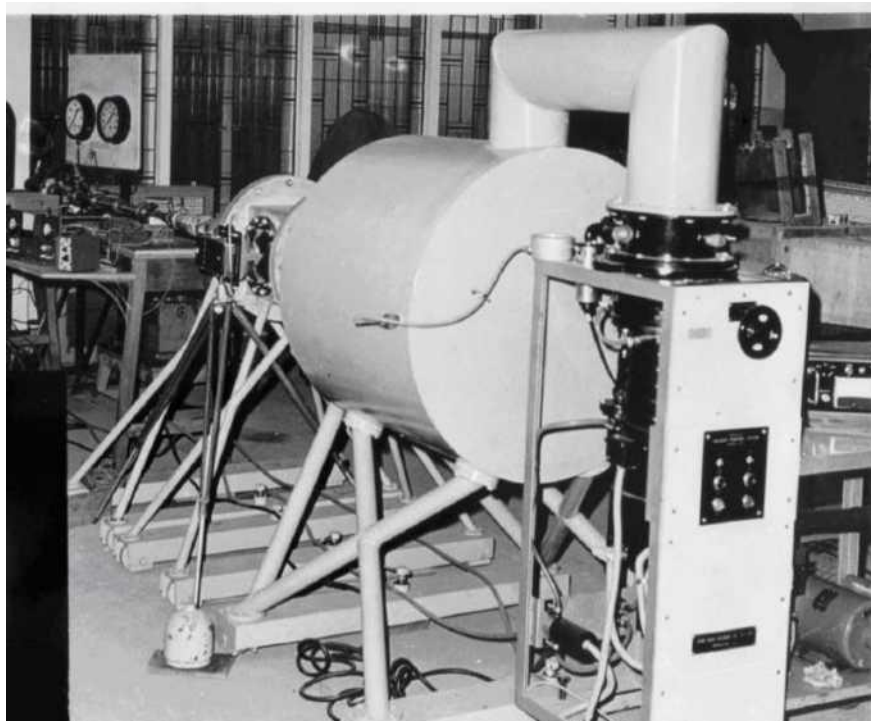
LABORATORY FOR HYPERSONIC
AND SHOCK WAVE RESEARCH

Lab Photo

Hypersonic Shock Tunnel 1

(*India's first Hypersonic Shock Tunnel*)

Driver Section Length	2 m
Driven Section Length	5 m
Internal Diameter	50 mm
External Diameter	101 mm
Test Section	300 mmx300 mmx450mm
Mach Number Range	4 to 13



Hypersonic Shock Tunnel 2

Driver Section Length	2m
Driven Section Length	5.12m
Internal Diameter	50mm
External Diameter	75mm
Test Section	300mmx300mmx450mm
Mach Number Range	5.75 to 12
Maximum Enthalpy	0.7 to 3 MJ/kg
Reynold’s Number Range	1 to 2 million/m

Hypersonic Sock Tunnel 3

(Free Piston Shock Tube)

Reservoir Length	1.3m
Compression tube length	10m
Driven tube length	4.5m
Piston mass	20 to 30 kg
Compression ratio	60
Test Section	300mmx300mmx450mm
Mach Number Range	6 to 12
Maximum Enthalpy	3 to 25 MJ/kg
Reynold's Number Range	1 to 2 million/m

Hypersonic Shock Tunnel 4

Driver Section Length	2m
Driven Section Length	10m
Internal Diameter	165mm
External Diameter	200mm
Test Section	1mx1mx2.85m
Mach Number Range	6 to 15
Maximum Enthalpy	8 MJ/kg
Reynold’s Number Range	1 to 2 million/m

Hypersonic Shock Tunnel 5

(Combustion driven)

Driver Section Length	3m
Driven Section Length	4.7m
Internal Diameter	102mm
External Diameter	130mm
Test Section	300mmx300mmx
Mach Number Range	6 to 13
Maximum Enthalpy	2 to 10 MJ/kg
Reynold's Number Range	1 to 2 million/m

Combustion Driven mode – 4 circumferential spark plugs 2cm upstream of diaphragm station

Chemical Shock Tunnel 1

Driver Section Length	1.8 m
Driven Section Length	3.3 m
Internal Diameter	50.8 mm
External Diameter	100 mm
Test Section	0.4 m

Chemical Shock Tunnel 3

Driver Section Length	3.6
Driven Section Length	7.6
Internal Diameter	54 mm
External Diameter	100 mm
Test Section	0.5 m

Flow Reactor

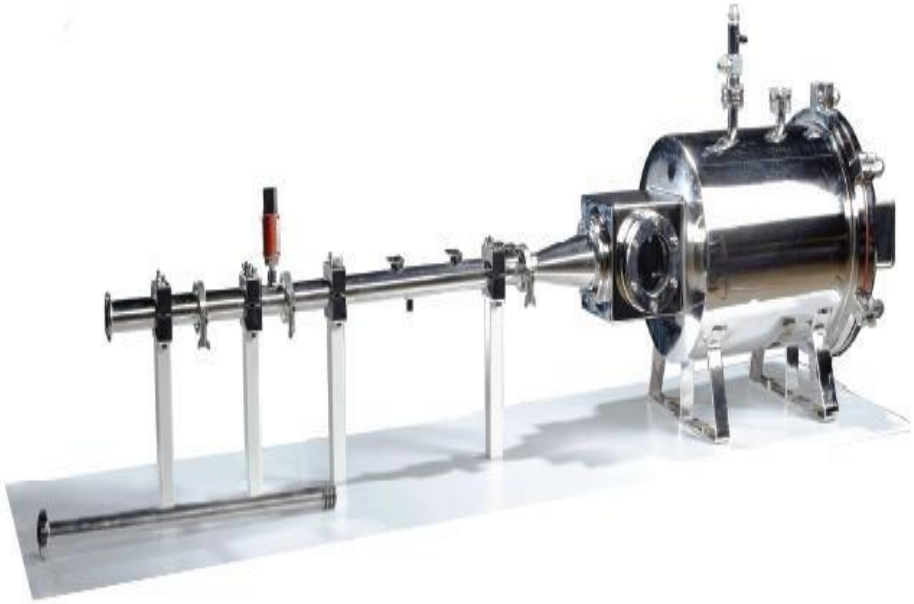
VST

Driver Section	1.5 m
Driven Section	4.3 m
Internal Diameter	136 mm
External Diameter	168 mm
Shock Pressure	100 bar
Test Section	870mm x 150mm x 500mm
Decay time	1.6 to 20 ms

Sputtering unit

Reddy Tunnel

Driver Section Length	0.4 m
Driven Section Length	0.6 m
Internal Diameter	29mm
Test Section	up to 50 mm (cross-wise dimension)
Mach Number Range	4 to 13
Maximum Enthalpy	2 MJ/kg



Co2 Facility